

N6Cam — Test Report

Date: 2026-07-07 20:14:25
Device: SIANA N6Cam (STM32N657 / IMX335 / Neural-ART NPU)
Firmware: 01.08.2593089169
MCU UID: 003031564236500400420024
Shell TTY: /dev/ttyACM0
Runtime: 56 seconds

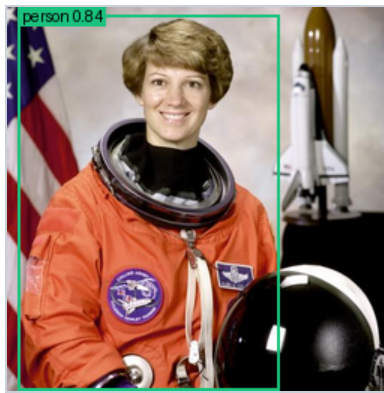
Result: ALL TESTS PASSED — Total: 65 | Pass: 65 | Fail: 0 | Skip: 0

Test ID	Description	Result
T00.1	Kit CDC port /dev/ttyACM0 exists and is readable	PASS
T00.2	Shell `help` lists user-defined commands	PASS
T00.3	`fw` returns FW VER: ...	PASS
T00.4	`uid` returns MCU UID	PASS
T00.5	`uptime` returns UPTIME ms	PASS
T01.1	`echo query` reports state + ok ack	PASS
T01.2	`echo off` succeeds (ok-ack)	PASS
T01.3	After `echo off`, no echoed prompt before reply	PASS
T01.4	`echo on` restores echo + ok ack	PASS
T02.1	`rtc` returns DD/MM/YY HH:MM:SS	PASS
T02.2	`rtc set DDMMYYYYHHMMSS` accepts a valid value	PASS
T02.3	RTC reads back the value we just set	PASS
T02.4	`version` prints Application:	PASS
T03.1	`irled on` reports state=1 + ok-ack	PASS
T03.2	`irled off` reports state=0	PASS
T03.3	`motion sense 65 45` accepted, echoed back	PASS
T03.4	`motion query` returns persisted 65/45	PASS

T04.1	`img quality 75` set + ok-ack	PASS
T04.2	`img color RGB` accepted	PASS
T04.3	`img chroma 1` accepted	PASS
T04.4	`img query` shows quality=75 color=RGB chroma=1	PASS
T05.1	`detect start` succeeds	PASS
T05.2	`detect profile 3 1` (people+vehicles, save-SD)	PASS
T05.3	`detect profile query` returns 0x03 / 0x01	PASS
T06.1	`notify enable 0x10` (people-detect bit) accepted	PASS
T06.2	`notify period 30` accepted	PASS
T06.3	`notify query` shows mask=0x10 period=30	PASS
T06.4	`notify trigger 16` emits +SDVRNTF JSON with rsn=16	PASS
T06.5	`notify disable` succeeds	PASS
T07.1	`sd query` reports mounted	PASS
T07.2	`sd ls` returns at least one entry	PASS
T07.3	`photo savesd` returns SoW §7 filename 4325412_07072026_201337.rdy	PASS
T07.4	New .rdy file appears on SD after `photo savesd` 4325412_07072026_201337.rdy	PASS
T08.1	`frame clear` reverts to live camera	PASS
T08.2	Upload a synthetic 192x192 grey frame	PASS
T08.3	`frame query` reports loaded	PASS
T08.4	`frame run` returns a count + NN time on a grey frame count=0 NN=88.1ms	PASS
T09.1	astronaut.jpg (real person) → NN detects ≥1 person count=1 NN=88.1ms	PASS



input (256×256)



kit detections

camera.jpg (real person) → NN detects ≥ 1 person
count=1 NN=88.1ms

T09.2



input (256×256)

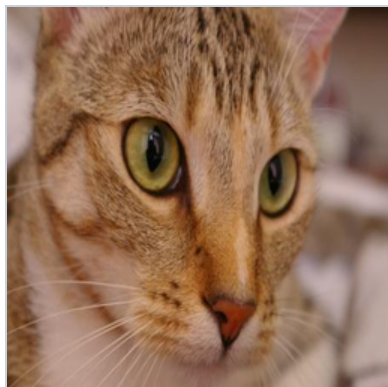


kit detections

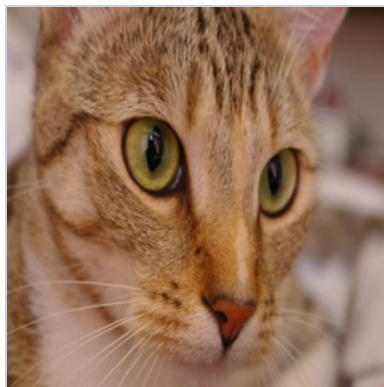
PASS

cat.jpg (no person) → NN returns 0 detections
count=0 NN=88.1ms

T09.3



input (256×256)



kit detections

PASS

T09.4 coffee.jpg (no person) → NN returns 0 detections
count=0 NN=88.1ms

PASS

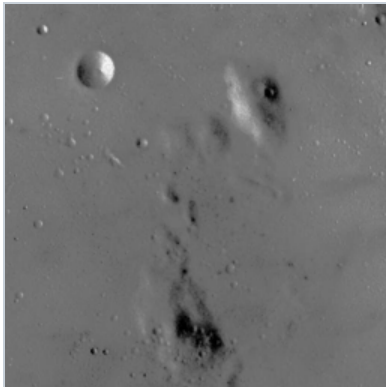


input (256×256)

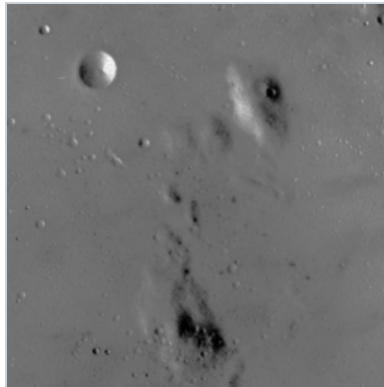


kit detections

moon.jpg (no person) → NN returns 0 detections
count=0 NN=88.1ms



input (256×256)

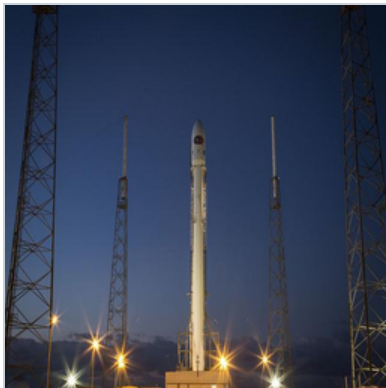


kit detections

T09.5

PASS

rocket.jpg (no person) → NN returns 0 detections
count=0 NN=88.1ms



input (256×256)



kit detections

T09.6

PASS

T09.7 NN inference avg ≤ 150 ms (SoW ≥5 FPS) over 6 images
avg=88.1ms worst=88.1ms n=6

PASS

T10.1 `camera awb auto` accepted

PASS

T10.2 `camera exposure 10000` accepted

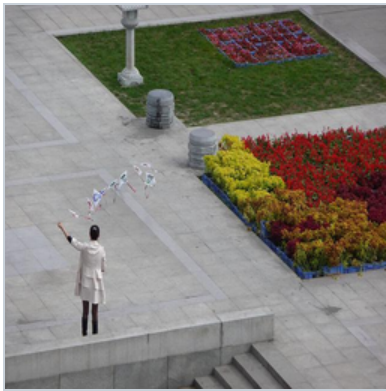
PASS

T10.3 `camera gain 0` accepted

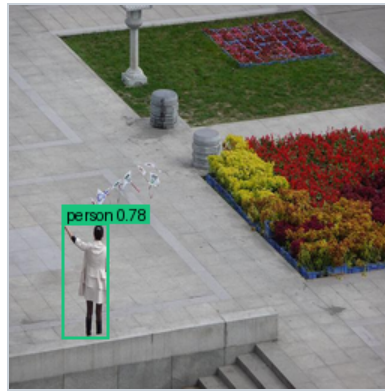
PASS

T11.1 1_person.jpg (single person) → ≥1 person
detected: person=1 vehicle=0 (gt 1p/0v) NN=88.1ms

PASS



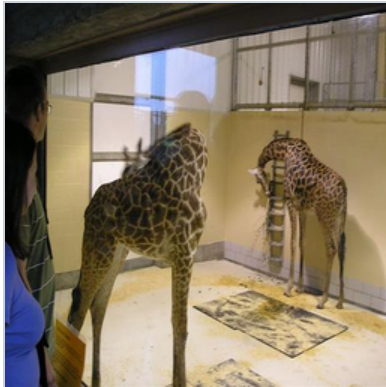
input (256×256)



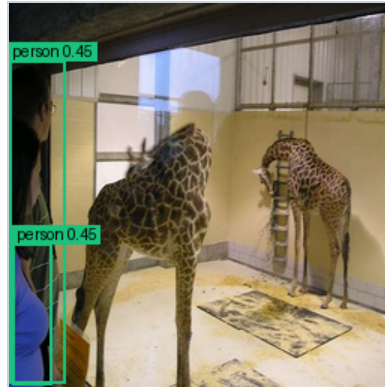
kit detections

2_people.jpg (small group) → ≥1 person
detected: person=2 vehicle=0 (gt 2p/0v) NN=88.1ms

T11.2



input (256×256)



kit detections

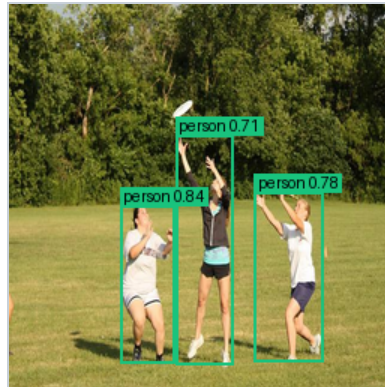
PASS

3_people.jpg (small group) → ≥2 person
detected: person=3 vehicle=0 (gt 3p/0v) NN=88.1ms

T11.3



input (256×256)



kit detections

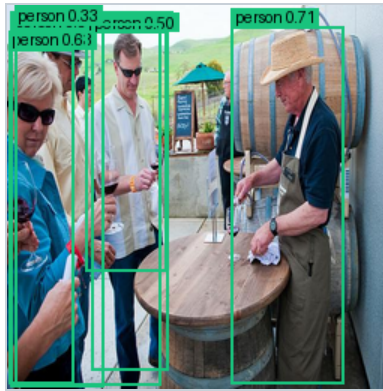
PASS

T11.4 7_people.jpg (group) → ≥2 person
detected: person=6 vehicle=0 (gt 7p/0v) NN=88.6ms

PASS



input (256×256)



kit detections

cars_real.jpg (cars + people) → ≥1 person
detected: person=1 vehicle=3 (gt 3p/3v) NN=88.7ms

T11.5



input (256×256)



kit detections

PASS

cars_real3.jpg (cars only) → ≥0 person
detected: person=0 vehicle=7 (gt 0p/6v) NN=88.6ms

T11.6



input (256×256)



kit detections

PASS

T11.7 13ppl_5trucks.jpg (mixed cars+people) → ≥3 person
detected: person=7 vehicle=3 (gt 13p/8v) NN=88.6ms

PASS



input (256×256)



kit detections

cars_15.jpg (vehicle: trains (not cars)) → ≥0 person
detected: person=0 vehicle=1 (gt 0p/15v) NN=88.7ms

T11.8



input (256×256)



kit detections

PASS

cars_18.jpg (vehicle: airplane (not cars)) → ≥0 person
detected: person=0 vehicle=1 (gt 0p/18v) NN=88.6ms

T11.9



input (256×256)



kit detections

PASS

T12.1 `tile default` restores 5x4/576 factory settings

PASS

T12.2 `tile query` reports frame/grid/crop geometry

PASS

T12.3 `tile grid 2 2` persists (query reflects it)

PASS

T12.4 `tile upload` accepts a 512x512 frame (CRC ok)

PASS

T12.5	<code>`tile run`</code> reports tiles + raw/thresh/NMS + timing tile run: 4 tiles (2x2) crop 384 over 512x512, 4 raw -> 2 over-thresh -> 2 after NMS, 75	PASS
T12.6	tilerun detects ≥ 1 person in 5_people.jpg persons=2	PASS
T12.7	NMS is non-expanding (raw $\geq$ over-thresh $\geq$ after-NMS) raw=4 thresh=2 nms=2	PASS
T12.8	<code>`tile live`</code> sweeps the live camera and returns a report tile live: 6 tiles (3x2) crop 400 over 800x600, 2 raw -> 0 over-thresh -> 0 after NMS, 1	PASS
T12.9	<code>`tile default`</code> restores after live + kit healthy	PASS